

Module 1 – Node.js Fundamentals

1. Intro to Node.js

What is Node.js?

Node.js is a JavaScript runtime built on Chrome's V8 engine that allows JavaScript to run outside the browser.

Why Node.js?

- Fast and scalable
 - Event-driven architecture
 - Uses JavaScript on backend
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2. Node Architecture

What is Node Architecture?

Node.js uses a single-threaded, non-blocking event loop model.

Key Concepts

- Event-driven system
 - Non-blocking I/O
 - Handles multiple requests efficiently
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3. Modules & NPM

What are Modules?

Modules are reusable blocks of code in Node.js.

Types of Modules

- Built-in modules (fs, http, path)
 - Custom modules
 - Third-party modules
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What is NPM?

NPM (Node Package Manager) is used to install and manage libraries.

Example:

```
const fs = require("fs");
```

4. File System Module

What is FS Module?

The File System module is used to work with files in Node.js.

Common Operations

- Read files
- Write files
- Update files
- Delete files

Example:

```
const fs = require("fs");  
  
fs.writeFileSync("test.txt", "Hello Node.js");
```

5. Event Loop

What is Event Loop?

Event Loop is the core mechanism that handles asynchronous operations in Node.js.

How It Works

- Executes synchronous code first
- Handles async tasks later
- Uses callback queue

Key Points

- Non-blocking behavior
 - Improves performance
 - Handles multiple requests
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6. Key Concepts to Remember

- Node.js runs JavaScript outside browser
- Uses event-driven architecture

- Modules help organize code
 - NPM manages packages
 - File system handles file operations
 - Event loop manages async tasks
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7. Summary

Node.js fundamentals provide the foundation for backend development. Understanding runtime, modules, file system, and event loop is essential for building efficient and scalable server-side applications.